

JEE(Main+Advanced) : ENTHUSIAST COURSE [PHASE : S, I(A), I, LIVE-I(A) & LIVE-I]**ANSWER KEY****PAPER-1****PART-1 : PHYSICS**

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B	C	B	D	C	A,B,C,D	B,C,D	A,D	A,B,C	B,C
SECTION-III	Q.	11	12	13							
	A.	B,C	A,B,D	A,D							

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A	C	A	D	D	A,B	B,C,D	A,C	A,B,C	A,B,C
SECTION-III	Q.	11	12	13							
	A.	A,B,D	A,C	A,B,C							

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A	D	B	C	D	A,C,D	B,D	B,D	A,C,D	B,C
SECTION-III	Q.	11	12	13							
	A.	B,D	B,D	A,B,C,D							

JEE(Main+Advanced) : ENTHUSIAST COURSE [PHASE : S, I(A), I, LIVE-I(A) & LIVE-I]**ANSWER KEY****PAPER-2****PART-1 : PHYSICS**

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	C	A	B	B	C	C	A,C,D	A,C,D	A,B,C	A,B,C,D
SECTION-III	Q.	11	12	13	14	15	16	17	18		
	A.	A,B	B,D	A,C,D	C,D	B	C	C	A		

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	C	C	A	B	C	B	A,B	A,C,D	A,C	A,B,C,D
SECTION-III	Q.	11	12	13	14	15	16	17	18		
	A.	B,D	A,C,D	A,C	B,C,D	C	C	A	C		

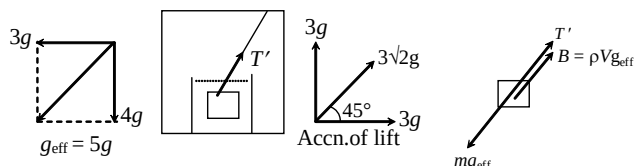
PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A	D	B	C	C	A	B,C	A,C	A,C	A,D
SECTION-III	Q.	11	12	13	14	15	16	17	18		
	A.	A,B,C,D	B,C,D	A,C	A,D	A	B	C	D		

PAPER-1**PART-1 : PHYSICS****SOLUTION****SECTION-I****1. Ans. (B)****Sol.** Let the volume of block is V .

$$T + \rho Vg = mg \Rightarrow 2l + \rho Vg = 30$$

$$\rho Vg = 9N$$



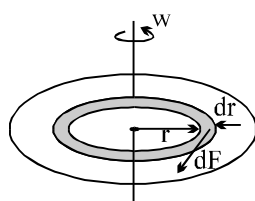
$$T + \rho Vg_{\text{eff}} = mg_{\text{eff}}$$

$$T' = 5mg - 5\rho Vg$$

$$= 5(mg - \rho Vg) = 105N$$

2. Ans. (C)**Sol.** Let coefficient of viscosity & thickness of film $\rightarrow \eta$ & x respectivelyViscous force on element ring is $dF = \eta$

$$(2\pi r dr) \left[\frac{\omega r - 0}{x} \right]$$



$$d\tau = (dF) r$$

$$\therefore \tau = \frac{\eta 2\pi \omega}{t} \int_0^R r^3 dr = \left(\frac{\eta (2\pi) \omega}{4x} \right) R^4$$

$$\therefore \tau \propto R^4$$

3. Ans. (B)

$$\text{Sol. } \omega_1 = \sqrt{\frac{k}{m_1}} = \sqrt{\frac{1800}{2}} = 30 \text{ rad/s}$$

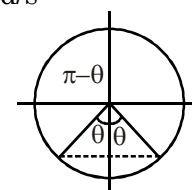
$$\omega_2 = \sqrt{\frac{1800}{8}} = 15 \text{ rad/s}$$

$$\pi - \theta = 15t$$

$$\pi + \theta = 30t$$

$$2\pi = (30 + 15)t$$

$$t = \frac{2\pi}{45} \text{ sec}$$

**4. Ans. (D)**

$$\text{Sol. } T_A = 2\pi \sqrt{\frac{I_A}{mgl}} T_B = 2\pi \sqrt{\frac{I_B}{mg(L-l)}}$$

Using of parallel axis theorem.

$$I_A = I_{cm} + ml^2, \quad I_B = I_{cm} + m(L-l)^2$$

$$\text{Since } T_A = T_B = T$$

$$\Rightarrow \frac{I_A}{mgl} = \frac{I_B}{mg(L-l)} \Rightarrow \frac{I_A}{l} = \frac{I_B}{(L-l)}$$

$$\Rightarrow \frac{I_{cm} + ml^2}{l} = \frac{I_{cm} + m(L-l)^2}{(L-l)}$$

$$\Rightarrow \frac{I_{cm}}{l} + ml = \frac{I_{cm}}{(L-l)} + m(L-l)$$

$$\Rightarrow \frac{I_{cm}(L-l-l)}{l(L-l)} = m(L-2l)$$

$$\Rightarrow I_{cm} = ml(L-l)$$

$$\Rightarrow T_A = T = 2\pi \sqrt{\frac{ml(L-l) + ml^2}{mgl}} = 2\pi \sqrt{\frac{L}{g}}$$

$$\Rightarrow T^2 = 4\pi^2 \frac{L}{g} \Rightarrow g = L \left(\frac{2\pi}{T} \right)^2$$

5. Ans. (C)

Sol. For tube B radius of curvature $> \frac{R}{2}$ and

for tube A radius of curvature = R
as air in the bubbles does not flow

$$\frac{4S}{R} = \frac{4S'}{R'} \quad [\text{Say } R' \text{ be radius of curvature of bubble of tube B}]$$

$$R' = \frac{RS'}{S} > \frac{R}{2}$$

$$\text{So, } S' > \frac{S}{2}$$

6. Ans. (A,B,C,D)

Sol. At $t = 0$

$$x_1 = 0$$

$$x_2 = 2\sqrt{3}$$

$$x = x_1 + x_2 = 2\sqrt{3} \text{ m.}$$

Resultant Amp.

$$A = \sqrt{(2)^2 + (4)^2 + 2(2)(4)\cos\left(\frac{\pi}{3}\right)}$$

$$A = \sqrt{28}$$

$$\text{Max. speed} = A\omega = 10\sqrt{28} = 20\sqrt{7} \text{ m/s}$$

$$\text{Max. acc.} = A\omega^2 = 100\sqrt{28} = 200\sqrt{7} \text{ m/sec}^2$$

$$\text{Energy} = \frac{1}{2} m\omega^2 A^2 = \frac{1}{2} (20 \times 10^{-3}) \times (10)^2$$

$$(\sqrt{28})^2 = 28 \text{ J}$$

7. Ans. (B,C,D)

$$\text{Sol. } F = -\frac{du}{dx}$$

$$F = -10 \times +20$$

$$F = 0 \text{ at } x = 2\text{m so } V \text{ is maximum at } x = 2\text{m}$$

$$T = 2\pi\sqrt{\frac{0.1}{10}} \quad \frac{\pi}{5} \text{ sec.}$$

8. Ans. (A,D)

$$\text{Sol. } v_0 = \omega\sqrt{A^2 - \left(\frac{A}{2}\right)^2}$$

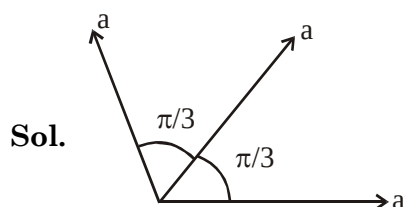
Velocity just after collision

$$= \frac{V_0}{2} = \sqrt{\frac{k}{2m}} \sqrt{A^2 - \left(\frac{A}{2}\right)^2}$$

$$\Rightarrow \frac{1}{2} \sqrt{\frac{k}{m}} \sqrt{\frac{3A^2}{4}} = \sqrt{\frac{k}{2m}} \sqrt{A^2 - \left(\frac{A}{2}\right)^2}$$

$$\Rightarrow A' = \sqrt{\frac{5}{8}} A$$

9. Ans. (A,B,C)



Sol.

(A) The vector sum of these three is $2a$

(C) Energy associated with each wave $\propto a^2$

Energy with resultant motion $\propto (2)^2 a^2$

(D) $V_{\text{max}} = A\omega$

$$V_{\text{initial}} = a\omega$$

$$V_{\text{resultant max.}} = 2a\omega = 2a\omega$$

10. Ans. (B,C)

Sol. Acceleration will be zero when

$$kx' = mg$$

$$300 \times x' = 1 \times 10$$

$$x' = \frac{1}{30} \text{ m}$$

For maximum compression x

$$mg(0.4 + x) - \frac{1}{2} kx^2 = 0$$

$$10(0.4 + x) - 150x^2 = 0$$

$$x = 0.2 \text{ m}$$

For maximum acceleration

$$= \frac{300(0.2) - 10 \times 1}{1} = 50 \text{ m/s}^2$$

11. Ans. (B,C)

Sol. (A) $3kx = 3mg$; equⁿ.

$$3k(x + x_0) - T - mg = ma \quad \dots(1)$$

$$T - 2mg = 2ma \quad \dots(2)$$

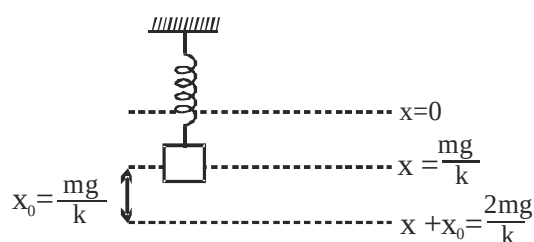
On solving,

$$T = 2k(x + x_0)$$

$$\therefore \text{for } x_0 = \frac{mg}{k}$$

$$T = 4mg$$

$$(D) \text{ If } x_0 = \frac{mg}{k}$$



If $x_0 > \frac{mg}{k}$ then string will become slack when 'm' comes to rest at top most extreme possible.

12. Ans. (A,B,D)

Sol. For ideal fluid flow

$$(\text{Volume flow rate})_A = (\text{Volume flow rate})_B$$

$$A_A V_A = A_B V_B$$

$$A_A > A_B \therefore V_B > V_A$$

$$\text{Also } P + \frac{1}{2}\rho V^2 = \text{constant}$$

$$V_B > V_A$$

$$P_B < P_A$$

13. Ans. (A,D)

Sol. Mass rate of strike $\frac{dm}{dt} = A(2v)\rho$

Change in velocity during strike

$$= \Delta v = \vec{v}_f - \vec{v}_i = (-v) - (v) = -2v$$

Thrust force on jet

$$= \Delta v \frac{dm}{dt} = \frac{dm}{dt} (-2v) = -4\rho A v^2$$

Thrust force on wall =

$$- \text{Thrust force on jet} = 4\rho A v^2$$

$$\frac{d(KE)}{dt} = \frac{1}{2} \frac{dm}{dt} (v_f^2) - \frac{1}{2} \frac{dm}{dt} (v_i^2)$$

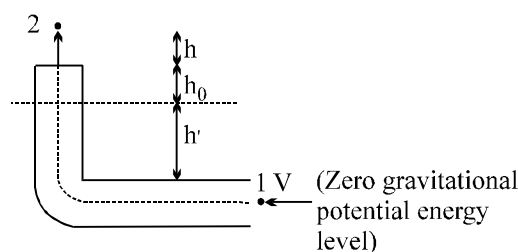
$$= \frac{1}{2} \frac{dm}{dt} [(-v)^2 - v^2] = 0$$

SECTION-III

1. Ans. 6

Sol. Applying Bernoulli's theorem for points 1 and 2 of the streamline shown

$P_0 \rightarrow$ Atmospheric pressure



$$\frac{1}{2}\rho V^2 + (P_0 + \rho gh') + 0$$

$$= 0 + P_0 + \rho g(h' + h_0 + h)$$

$$\Rightarrow h = 60 \text{ cm}$$

2. Ans. 5

Sol. Let, in equilibrium position, compression of spring be x . Liquid of volume $l^2 x$ is displaced from its original position and level of liquid in tank rises as shown in figure.

$$\text{This rise in level, } \Delta x = \frac{l^2 x}{A - l^2};$$

Where $A = 15 \text{ cm} \times 20 \text{ cm}$ (Base area of tank)

$$\text{or } \Delta x = 0.5x$$

$\therefore$ Mass of water displaced by the block $= l^2(x + \Delta x)\rho = 15x \text{ kg}$

upthrust exerted by water = apparent weight of water displaced,

$\therefore$ Upthrust

$$F_1 = 15x(g - a) = 120x \text{ Newton}$$

Upward force exerted by spring

$$F_2 = Kx = 280x$$

Considering free body diagram of the block, (Figure)

$$mg - (F_1 - F_2) = ma$$

Substituting values of F_1 and

$$F_2, x = 0.04m = 4 \text{ cm}$$

If the block is slightly pushed downward by dx , both F_1 and F_2 increase.

$$\text{Increase in } F_1 \text{ is } dF_1 = 120dx$$

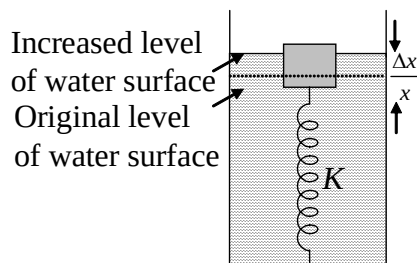
$$\text{Increase in } F_2 \text{ is } dF_2 = 280dx$$

Restoring force on block = increase in F_1 + increase in F_2

$$= dF_1 + dF_2 = (120dx + 280dx) = 400dx$$

or Restoring acceleration

$$= \frac{400 \cdot dx}{m} = 200 \cdot dx = \omega^2 (dx) \Rightarrow \omega = 10\sqrt{2}$$



Since, restoring acceleration $\propto$ displacement (dx)

Therefore, block performs SHM.

Hence, frequency,

$$f = \frac{5\sqrt{2}}{\pi} \text{ per second.}$$

$$a = 5$$

3 Ans. 1

Sol. If m is slightly displaced by x down words let elongation in spring is x_1

$$\text{Then } \frac{x_1}{b} = \frac{x}{a} \Rightarrow x_1 = \frac{b}{a} x$$

Let T = tension developed in string. Since rod is massless

$$T \cdot a = (KX_1)b$$

$$\Rightarrow T a = \left(K \cdot \frac{b}{a} x \right) b$$

$$\text{Time period} = 2\pi \sqrt{\frac{m}{Kb^2/a^2}} = 1$$

4. Ans. 8

Sol. Terminal velocity is given by

$$v = \frac{2r^2(\rho - \sigma)(g + a)}{9\eta}$$

$$= \frac{2 \times (0.05)^2 \times 10^{-6} \times (2000 - 1000)(9.8 + 1.2)}{9 \times 1.1 \times 10^{-3}}$$

$$= 5.55 \times 10^{-3} \text{ m/s}$$

Assuming that the sand particles reach their terminal velocity quickly, minimum

$$\text{time } t_{\min} = \frac{45.45 \times 10^{-3}}{5.55 \times 10^{-3}} = 8.18 \text{ s} \approx 8 \text{ s}$$

5. Ans. 1

Sol. $F_{\text{req}} = mg + 2[T(2\pi R)]$

$$[T = 75 \times 10^{-3} \text{ N/m}]$$

$$= 0.1 + 2 [75 \times 10^{-3} (0.2)] = 0.130 \text{ N}$$

PART-2 : CHEMISTRY**SOLUTION****SECTION-I**

1. Ans.(A)
2. Ans. (C)
3. Ans.(A)
4. Ans.(D)
5. Ans. (D)
6. Ans.(A,B)
7. Ans.(B,C,D)
8. Ans. (A, C)
9. Ans.(A,B,C)
10. Ans.(A,B,C)
11. Ans (A,B,D)
12. Ans.(A,C)

Sol. The distance between nearest neighbours is same in both forms means '2r' is same in both and hence the unit cell edge length, i.e., cell constant, a, is different in both.

$$a_{\alpha} = \frac{4r}{\sqrt{3}} \text{ and } a_{\beta} = \frac{4r}{\sqrt{2}}$$

The distance between next nearest neighbours in both form is 'a', which is different.

Void space in BCC = 32%, FCC = 26%

In both forms, the required diameter of largest spheres = (a - 2r) and 'a' is different

$$2r_c = \left(\frac{a\sqrt{3}}{4} \right)$$

13. Ans.(A,B,C)

SECTION-III

1. Ans.(7)
2. Ans.(7)

$$\text{Sol. P. Fr.} = \frac{8 \times \frac{4}{3} \pi r^3}{\left(\frac{8r}{\sqrt{3}} \right)^3} = \frac{8 \times \frac{4}{3} \times \frac{5.44}{\sqrt{3}}}{\left(\frac{8}{\sqrt{3}} \right)^3}$$

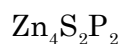
$$= \frac{5.44}{16} = 0.34$$

3. Ans.(3)
4. Ans. (5)
5. Ans.(8)

$$\text{Zn} = 4, \text{S}^{2-} \text{ (removed)} = 4 \times \frac{1}{2} = 2$$

$$\text{left S}^{-2} = 4 - 2 = (2)$$

Total positive charge that must be compensated is = +4.



PART-3 : MATHEMATICS

SOLUTION

SECTION-I

1. **Ans. (A)**

Sol. L.H.S is an even integer & R.H.S. is an odd integer.

2. **Ans. (D)**

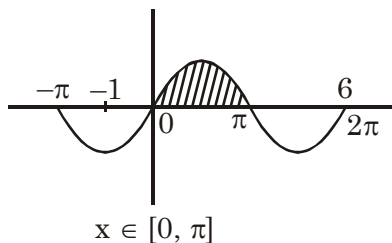
Sol. $3\sin x + 4\cos x = \frac{10y+2}{1-y}$

$$\Rightarrow -5 \leq \frac{10y+2}{1-y} \leq 5$$

$$\Rightarrow y \in \left[-\frac{7}{5}, \frac{1}{5}\right]$$

3. **Ans. (B)**

Sol. $\sin x \geq 0$ & $x \in [-1, 6]$



4. **Ans. (C)**

Sol. (A) $f(x) = |\sin x + \cos x|$

(B) $f(x) = \tan x$ where $2x \neq (2n+1)\frac{\pi}{2}$

$$(C) g(x) = \begin{cases} 1 & x > 0 \\ 0 & x = 0 = f(x) \\ -1 & x < 0 \end{cases}$$

(D) $D_{f(x)} : x \neq (2n+1)\frac{\pi}{2}$, $D_{g(x)} : x \neq n\frac{\pi}{2}$

5. **Ans. (D)**

Sol. Put $y = \frac{x}{2}$

$$\Rightarrow f\left(\frac{x}{2}\right) + f(x) + 8x + x^2 = f\left(\frac{x}{2}\right) + 2x^2 + 4x$$

$$\Rightarrow f(x) = x^2 - 4x$$

$$\Rightarrow f(5) = 5$$

6. **Ans. (A,C,D)**

Sol. Let $m_{AB} = m$, $m_{BE} = 2$

$$\frac{m-2}{1+2m} = \frac{2-1}{1+2} \Rightarrow m = 7$$

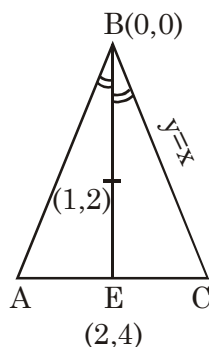
$\Rightarrow$ line AB is $y = 7x$

Let $A(\alpha, 7\alpha)$ & $C(\beta, \beta)$

$$\alpha + \beta = 4 \text{ \& } 7\alpha + \beta = 8$$

$$\Rightarrow \alpha = \frac{2}{3} \text{ \& } \beta = \frac{10}{3}$$

$$\Rightarrow A\left(\frac{2}{3}, \frac{14}{3}\right) \text{ \& } C\left(\frac{10}{3}, \frac{10}{3}\right)$$



7. **Ans. (B,D)**

Sol. Replace x by $x-2$

$$f(2+x) + f(2-x) = 0$$

$\Rightarrow$ Graph is symmetric about point $(2, 0)$

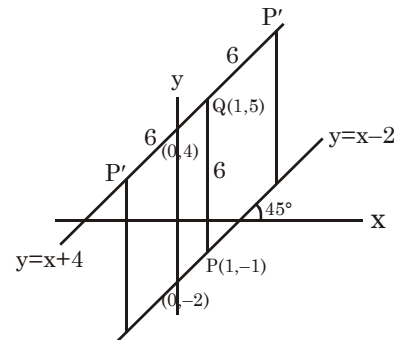
$f(x) = 0$ has one root 2 and other roots

are $2 \pm \alpha$, $2 \pm \beta$, $2 \pm \gamma$

$\Rightarrow$ sum of all roots = 14

8. **Ans. (B,D)**

Sol.



Let opposite vertex is $P'(t, t+4)$

$$\therefore QP' = 6$$

$$\therefore \sqrt{(t-1)^2 + (t+4-5)^2} = 6$$

$$\Rightarrow (t-1)^2 = 18 \text{ or } t = 1 \pm 3\sqrt{2}$$

$$\Rightarrow P'(1+3\sqrt{2}, 5+3\sqrt{2}) \text{ or } (1-3\sqrt{2}, 5-3\sqrt{2})$$

9. **Ans. (A,C,D)**

Sol. Angle bisectors of P_1 are $\frac{x^2-y^2}{2m} = \frac{xy}{-n}$
compare with P_2

$$\Rightarrow \frac{2m}{n} = -4n \text{ or } m = -2n^2$$

$$\text{put } x = 1 \text{ in } P_1 \Rightarrow m - 2ny - my^2 = 0$$

$$\text{intercept} = |\alpha - \beta| = \frac{\sqrt{D}}{|a|} = 2\sqrt{5}$$

$$\Rightarrow 4n^2 + 4m^2 = 20m^2$$

$$\Rightarrow -2m = 16m^2 \Rightarrow m = -\frac{1}{8} \text{ \& } n = \pm \frac{1}{4}$$

10. **Ans. (B, C)**

Sol. $\begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix} = 0$

$$\Rightarrow a = b = c \text{ (reject) or } a + b + c = 0$$

$$\Rightarrow 2^{a^2b^{-1}c^{-1}} \cdot 2^{b^2c^{-1}a^{-1}} \cdot 2^{c^2a^{-1}b^{-1}} = 2^{\frac{a^2}{bc} + \frac{b^2}{ca} + \frac{c^2}{ab}}$$

$$= 2^{\frac{a^3+b^3+c^3}{abc}} = 2^3 = 8$$

11. Ans. (B,D)

Sol. Let $f(x) - 2x - 1 = \lambda x(x-1)(x-3)$

put $x = 4$

$$33 - 9 = \lambda(4 \cdot 3 \cdot 1)$$

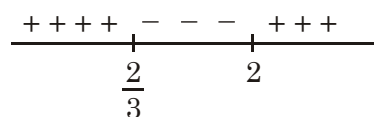
$$\lambda = 2$$

$$\Rightarrow f(x) = 2x(x-1)(x-3) + 2x + 1$$

$$= 2x^3 - 8x^2 + 8x + 1$$

$$f'(x) = 6x^2 - 16x + 8$$

$$= 2(3x-2)(x-2)$$



$f(x)$ is many one in $[0, 4]$

$$f(0) = f(2) = 1 \quad (\text{least value})$$

$$f(4) = 33 \quad (\text{greatest value})$$

$$\Rightarrow \text{range} = [1, 33]$$

12. Ans. (B,D)

Sol. $(2^y)^2 - x \cdot 2^y + 4 = 0$

$$2^y = \frac{x \pm \sqrt{x^2 - 16}}{2}$$

$$\Rightarrow y = f(x) = \log_2 \left(\frac{x + \sqrt{x^2 - 16}}{2} \right) \quad \{\because \text{Range is } [1, \infty)\}$$

for inverse $x \leftrightarrow y$

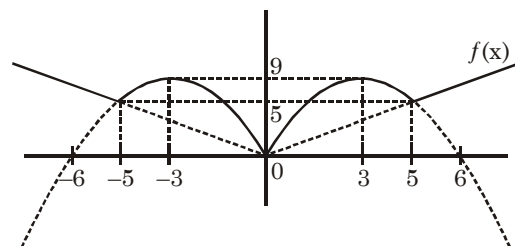
$$\Rightarrow 4^x + 4 = y \cdot 2^x$$

$$\Rightarrow y = 2^x + 4 \cdot 2^{-x} = f^{-1}(x)$$

$$f^{-1}(2) = 5 \text{ \& } f(5) = 2$$

13. Ans. (A,B,C,D)

Sol.



SECTION-III

1. Ans. 3

Sol. $f'(x) = 3x^2 + 2x + 1 > 0 \quad (\because D < 0)$

$\Rightarrow f(x)$ is increasing

$$\therefore f(x) = f^{-1}(x) \Rightarrow f(x) = x$$

$$x^3 + x^2 + x - 36 = x$$

$$\Rightarrow x^3 + x^2 - 36 = 0$$

$$\Rightarrow x = 3$$

2. Ans. 2

Sol. $\text{fog}(x) = \text{gof}(x)$

$$\Rightarrow \frac{10^{x+1} - 1}{10^{x+1} + 1} = \frac{10^x - 1}{10^x + 1} + 1 = \frac{2 \cdot 10^x}{10^x + 1}$$

use C & D

$$\Rightarrow 10^{x+1} = \frac{3 \cdot 10^x + 1}{1 - 10^x} \quad \text{Let } 10^x = t$$

$$\Rightarrow 10t^2 - 7t + 1 = 0 \Rightarrow t = \frac{1}{2} \text{ or } \frac{1}{5}$$

3. Ans. 3

Sol. Combined equation of lines is

$$(ax + y + 1)(x + by) = 0$$

homogenize -

$$ax^2 + by^2 + (ab + 1)xy + (x + by)(ax + by) = 0$$

$$\text{perpendicular lines} \Rightarrow a + a + b + b^2 = 0$$

$$\Rightarrow 2a = -b - b^2 = \frac{1}{4} - \left(b + \frac{1}{2}\right)^2 \leq \frac{1}{4}$$

$$\Rightarrow 24a \leq 3$$

4. Ans. 6

$$\text{Sol. } f(x) = 4 \frac{\sqrt{(1 - \sqrt{x-1})^2}}{2-x} = 4 \frac{|1 - \sqrt{x-1}|}{2-x}$$

$$= \begin{cases} 4 \cdot \frac{1 - \sqrt{x-1}}{2-x} = \frac{4}{1 + \sqrt{x-1}} & 1 \leq x < 2 \\ 4 \cdot \frac{\sqrt{x-1} - 1}{2-x} = \frac{-4}{1 + \sqrt{x-1}} & x > 2 \end{cases}$$

if $x \in [1, 2)$, then $y \in (2, 4]$

if $x \in [2, \infty)$, then $y \in (-2, 0)$

$$\text{sum} = 3 + 4 - 1 = 6 \text{ Ans.}$$

5. Ans. 2

$$\text{Sol. } 1009 \ell = \sum_{n=1}^{2020} [nx] = \sum_{n=1}^{2020} [(2021-n)x]$$

$$\sum_{n=1}^{2020} [1010 - nx]$$

$$\text{Also } [nx] + [1010 - nx] = 1010 + [nx] + [-nx] \\ = 1009 \quad (nx \neq I)$$

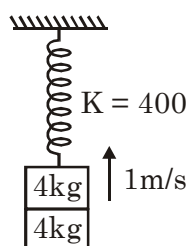
$$(2 \times 1009)\ell = \sum_{n=1}^{2020} 1009 \Rightarrow \ell = 1010$$

PAPER-2**PART-1 : PHYSICS****SOLUTION****SECTION-I**1. **Ans. (C)**

Sol. The only force acting on the body is the viscous force

$$\text{Here, } m \frac{v dv}{dx} = -6\pi\eta r v = -rv$$

$$\Rightarrow \int_v^0 m dv = \int_0^x -r dx \Rightarrow x = \frac{mv}{r}$$

2. **Ans. (A)****Sol.**

$$\omega = \sqrt{\frac{K}{m}} = \sqrt{\frac{400}{8}} = \sqrt{50}$$

$$\omega^2 = 50$$

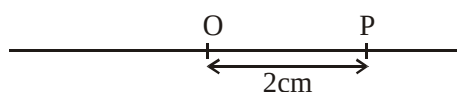
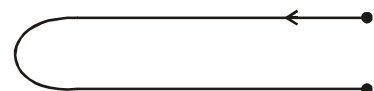
$$V = \omega \sqrt{A^2 - x^2}$$

$$V^2 = \omega^2 (A^2 - x^2)$$

$$1 = 50 \left(A^2 - \frac{1}{100} \right)$$

$$A^2 = \frac{1}{50} + \frac{1}{100}$$

$$A^2 = \frac{3}{100}$$

3. **Ans. (B)****Sol.**

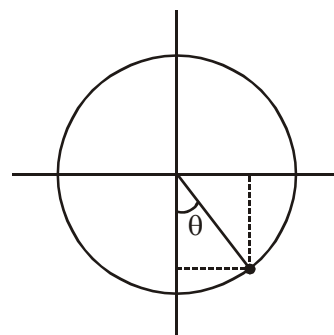
$$v^2 = \omega^2 (A^2 - x^2) \quad \omega = \frac{2\pi}{T} = \frac{1}{2}$$

$$1 = \frac{1}{4} (A^2 - (2)^2)$$

$$A = 2\sqrt{2}$$

So time from P to O

Taking projection on horizontal axis



$$2\sqrt{2} \sin \theta = 2$$

$$\theta = 45^\circ \Rightarrow \frac{\pi}{4}$$

So total angle rotated by phasor

$$= \pi + \frac{\pi}{4} + \frac{\pi}{4} = \frac{3\pi}{2}$$

$$t = \frac{\frac{3\pi}{2}}{\frac{1}{2}} = 3\pi \text{ sec}$$

4. **Ans. (B)****Sol.** Torque on gate due to fluid

$$= \int (\rho g x) (B dx) x$$

$$= \rho g B \int_0^L x^2 dx \Rightarrow \frac{\rho g B L^3}{3}$$

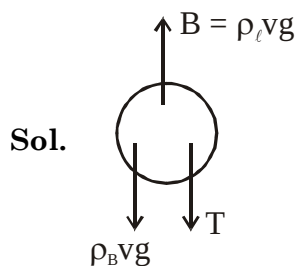
Torque due to Force = (F) (L - 1)

$$\therefore F(L-1) = \frac{\rho g B L^3}{3}$$

$$F = \frac{\rho g B L^3}{3(L-1)}$$

Where B = Breadth = 4m, L = length = 4m

5. Ans. (C)



In frame of container $g_{\text{eff}} = 12 \text{ m/s}^2$

$$v = \frac{M}{\rho_B}$$

$$T = (\rho_\ell - \rho_B)vg = kx$$

$$x = \frac{(\rho_\ell - \rho_B)vg}{k}$$

$$= \left(\frac{\rho_\ell}{\rho_B} - 1 \right) \frac{Mg}{k} = \frac{0.1 \times 10 \times 12}{200} \text{ m}$$

$$= 6 \text{ cm}$$

6. Ans. (C)

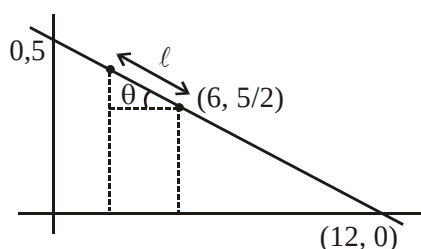
Sol. $g_{\text{eff}} = \sqrt{g^2 + \left(\frac{g}{2}\right)^2 + 2(g)\left(\frac{g}{2}\right)\cos(60^\circ)}$

$$g_{\text{eff}} = 13.23 \text{ m/s}^2$$

$$T = 2\pi\sqrt{\frac{\ell}{g_{\text{eff}}}}$$

7. Ans. (A,C,D)

Sol. Amplitude = $\frac{13}{2} \text{ m}$, $\omega = \frac{2\pi}{T} = \frac{\pi}{4} \text{ s}^{-1}$



$$x = (6 - \ell \cos \theta), \quad y = \frac{5}{2} + \ell \sin \theta$$

$$\ell \Rightarrow \frac{13}{2} \cos(\omega t), \quad \sin \theta = \frac{5}{13}, \cos \theta = \frac{12}{13}$$

$$\vec{r} = \frac{13}{2}(1 - \cos \omega t) \frac{12}{13} \hat{i} + \left[5 - \frac{13}{2}(1 - \cos \omega t) \frac{5}{13} \right] \hat{j}$$

$$\therefore \vec{r} = 6(1 - \cos \omega t) \hat{i} + \frac{5}{2}(1 + \cos \omega t) \hat{j}$$

$$\vec{r} = 12 \sin^2 \left(\frac{\pi t}{8} \right) \hat{i} + 5 \cos^2 \left(\frac{\pi t}{8} \right) \hat{j}$$

$$\therefore \vec{v} = \left[\frac{3\pi}{2} \sin \left(\frac{\pi t}{4} \right) \hat{i} - \frac{5\pi}{8} \sin \left(\frac{\pi t}{4} \right) \hat{j} \right] \text{ m/s}$$

8. Ans. (A,C,D)

Sol. $T = 0.2 = 2\pi\sqrt{\frac{m}{k}}$

$$\Rightarrow m = 0.2 \text{ kg}$$

At $t = 0$, particle is at extreme position and moving towards +ve x-axis

$$K_{\text{max}} = \frac{1}{2} m (v_{\text{max}})^2 = 4.0 \text{ J}$$

9. Ans. (A,B,C)

Sol. Let acceleration of ball relative to the cube

$$\text{is } \vec{a}_{b,c} = a'_x \hat{i} + a'_y \hat{j} + a'_z \hat{k}$$

Now,

$$a'_x = \left(\frac{\rho - \frac{\rho}{3}}{\frac{\rho}{3}} \right) a_x = 2a_x = 2 \times \frac{5}{\sqrt{2}} = 5\sqrt{2} \text{ m/s}^2$$

$$a'_y = \left(\frac{\rho - \frac{\rho}{3}}{\frac{\rho}{3}} \right) (g + a_y) = 2(g + a_y) = 2 \left(10 - \frac{15}{2} \right) = 5 \text{ m/s}^2$$

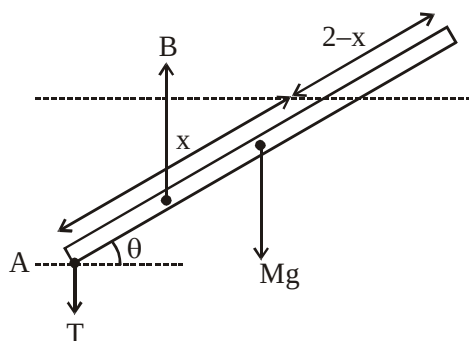
$$a'_z = \left(\frac{\rho - \frac{\rho}{3}}{\frac{\rho}{3}} \right) a_z = 2a_z = 2 \times \frac{5}{2} = 5 \text{ m/s}^2$$

$\therefore$ Acceleration of ball relative to the cube

$$\text{is } \vec{a}_{b,c} = (5\sqrt{2}\hat{i} + 5\hat{j} + 5\hat{k}) \text{ m/s}^2$$

10. Ans. (A,B,C,D)

Sol.



$$B = \rho_w (xA)g$$

$$mg = (0.25 \rho_w) (2A) g$$

Net torque about 'A' is zero.

$$\therefore \frac{x}{2} (\rho_w xA) g \cos \theta = (0.25 \rho_w) (2A) g (1) \cos \theta$$

$$\frac{x^2}{2} = \frac{1}{2} \Rightarrow x = 1$$

$$T = B - mg = \frac{1}{2} \rho_w Ag = 5$$

11. Ans. (A,B)

Sol. Level will fall if initially the impurity pieces were floating along with ice and later it sinks.

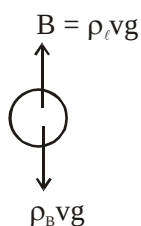
Level will remain unchanged if initially they were floating and later also they keep floating.

12. Ans. (B,D)

Sol. Speed of 2nd object just before entering

$$\text{liquid is } \sqrt{2 \times 10 \times 45} \Rightarrow 30 \text{ m/s}$$

FBD in liquid



$$a \Rightarrow \frac{(\rho_l - \rho_B)}{\rho_B} g$$

$$a = 4g$$

In liquid object will stop at

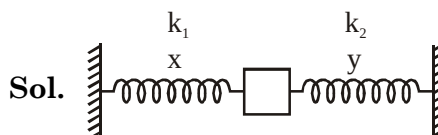
$$v^2 - u^2 = 2as$$

$$s = \frac{(30)^2 - 0^2}{2 \times 4 \times 10}$$

$$\Rightarrow 11.25 \text{ m}$$

$\therefore$ least distance will be $20 - 11.25 = 8.75 \text{ m}$

13. Ans. (A,C,D)



Sol.

x & y are extension in the spring

For eq.

$$k_1 x = k_2 y \quad \dots (1)$$

$$x + y = 0.2 \text{ m} \quad \dots (2)$$

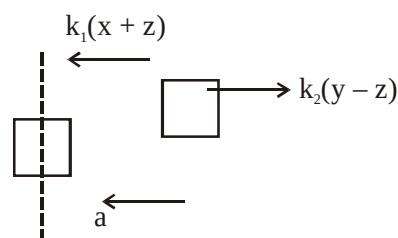
$$y \Rightarrow \frac{0.2}{4} \text{ m}, \quad x = \frac{0.6}{4} \text{ m}$$

$$= 5 \text{ cm}$$

$$= 15 \text{ cm}$$

For time period

Displace block by small distance let's say 'z'



$$a \Rightarrow \frac{k_1(x+z) - k_2(y-z)}{m} = \frac{(k_1 + k_2)z}{m}$$

$$\therefore \omega = \sqrt{\frac{k_1 + k_2}{m}}$$

$$\text{Potential energy} = \frac{1}{2} k_1 x^2$$

$$= \frac{1}{2} (1000) \left(\frac{15}{100} \right)^2$$

14. Ans. (C,D)

Sol. Average velocity in 1 cycle = 0

$$\Delta KE \text{ in 1 cycle} = 0$$

15. Ans. (B)

Sol. $V_{\max} = \frac{2}{3}\sqrt{gL}$ and $\omega = \sqrt{\frac{k}{m}} = 2\sqrt{\frac{g}{L}}$

$$\therefore A = \frac{V_{\max}}{\omega} = \frac{3}{4}L$$

Hence time taken t , from state of compression till block reaches mean position is given by

$$x = A \sin \omega t_0 \text{ where } x = \frac{L}{4}$$

$$\therefore t_0 = \sqrt{\frac{L}{4g}} \sin^{-1} \frac{1}{3}$$

Time taken by block to reach from mean position to bottom most position is

$$\frac{2\pi}{4\omega} = \frac{\pi}{4} \sqrt{\frac{L}{g}}$$

Hence the required time

$$= \frac{\pi}{4} \sqrt{\frac{L}{g}} + \sqrt{\frac{L}{4g}} \sin^{-1} \frac{1}{3}$$

16. Ans. (C)

Sol. At time $\frac{T}{2} + 2t_0$ block will reach at the same point where it was at $t = 0$.

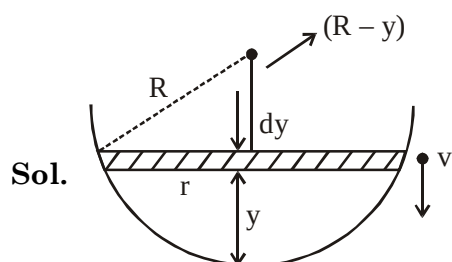
17. Ans. (C)

Sol. Velocity of liquid while coming out of hole

$$= \sqrt{(2g)\left(\frac{h}{2}\right)}$$

$$\therefore \text{Rate of volume outflow} = a\sqrt{gh}$$

18. Ans. (A)



Sol.

$$R^2 - (R - y)^2 = r^2$$

$$R^2 - [R^2 + y^2 - 2Ry] = r^2$$

$$2Ry - y^2 = r^2$$

$$r = \sqrt{2Ry - y^2}$$

$$a\sqrt{2gy} = v_1 \pi (2Ry - y^2)$$

$$v_1 = \frac{a}{\pi} \frac{\sqrt{2gy}}{(2Ry - y^2)} = -\frac{dy}{dt}$$

$$\sqrt{2g} \frac{a}{\pi} dt \Rightarrow -\frac{2Ry - y^2}{\sqrt{y}} dy$$

$$\Rightarrow 2R \int_0^R y^{1/2} dy - \int_0^R y^{3/2} dy$$

$$\Rightarrow \frac{4R^{5/2}}{3} - \frac{2}{5} R^{5/2}$$

$$\sqrt{2g} \frac{a}{\pi} t \Rightarrow \frac{14R^{5/2}}{15}$$

$$t \Rightarrow \frac{14\pi R^{5/2}}{a15\sqrt{2g}}$$

PART-2 : CHEMISTRY

SOLUTION

SECTION-I

1. Ans.(C)

2. Ans.(C)

3. Ans.(A)

$$\text{Sol. } 3.6 = \frac{4 \times \frac{20}{6 \times 10^{23}}}{\left[2 \left(x + \frac{302}{3} \right) \times 10^{-10} \text{ cm} \right]^3}$$

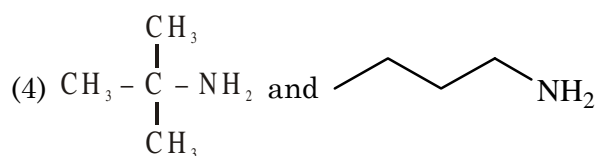
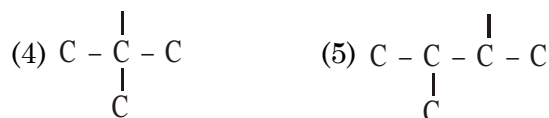
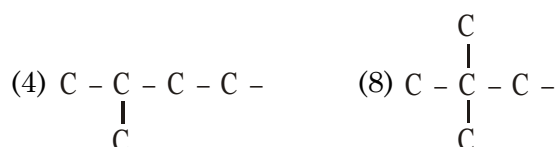
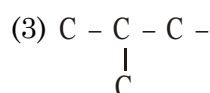
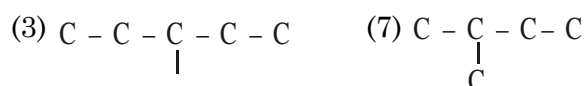
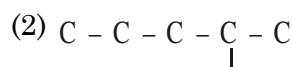
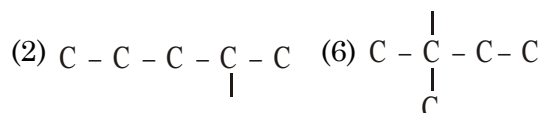
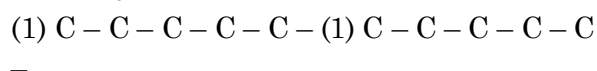
$$\Rightarrow x = 66 \text{ pm}$$

4. Ans.(B)

5. Ans. (C)

Sol. (2) C_5H_{11} has eight alkyl groups

(3) C_4H_9 has four alkyl groups



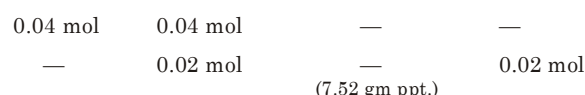
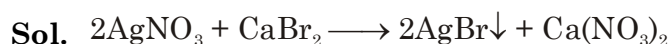
are having different pattern of Carbon skeleton.

6. Ans.(B)

7. Ans.(A,B)

8. Ans. (A, C, D)

9. Ans.(A,C)



$$\% \frac{w}{v} \text{ of Br}^- = \frac{0.02 \times 80 \times 2}{95} \times 100$$

$$\% \frac{w}{w} \text{ of Br}^- = \frac{0.02 \times 80 \times 2}{122.48} \times 100$$

10. Ans.(A,B,C,D)

11. Ans. (B, D)

12. Ans.(A,C,D)

13. Ans.(A,C)

14. Ans.(B,C,D)

15. Ans.(C)

Sol. $\rho = \frac{2 \times \frac{450}{6 \times 10^{23}}}{(10 \times 10^{-8} \text{ cm})^3} = 1.5 \frac{\text{gm}}{\text{cc}}$

16. Ans.(C)

Sol. $\frac{68}{1.5} = \frac{\text{P.Fr.}}{1.2} \Rightarrow \text{P.Fr.} 54.4\%$

$$\text{void} = 45.6\%$$

17. Ans. (A)

18. Ans. (C)

PART-3 : MATHEMATICS

SOLUTION

SECTION-I

1. **Ans. (A)**

Sol. $\log_5(x-1) \geq 0$

$$\Rightarrow x \in [2, \infty) \quad \dots (1)$$

$$4x + 5 - x^2 \geq 0$$

$$\Rightarrow x^2 - 4x - 5 \leq 0$$

$$x \in [-1, 5] \quad \dots (2)$$

$$(1) \cap (2) \text{ gives}$$

$$x \in [2, 5]$$

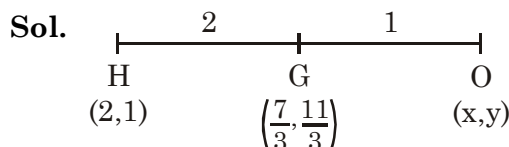
2. **Ans. (D)**

Sol. $f(x)$ is either increasing or decreasing but $f(0) = 5$

$$\Rightarrow f(x) \text{ is decreasing}$$

$$\text{Hence } f(3) = 0$$

3. **Ans. (B)**



$$\left(\frac{2x+2}{3}, \frac{2y+1}{3}\right) = \left(\frac{8}{3}, \frac{11}{3}\right)$$

$$x = 3 \text{ \& } y = 5$$

$$0 \equiv (3, 5)$$

4. **Ans. (C)**

Sol. $6\{x\} - 8\{x\}^2 - 1 \geq 0$

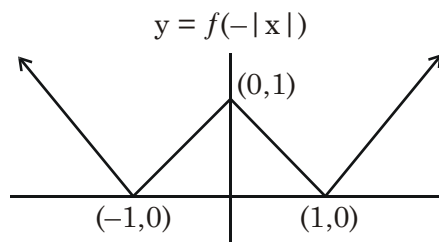
$$8\{x\}^2 - 6\{x\} + 1 \leq 0$$

$$\frac{1}{4} \leq \{x\} \leq \frac{1}{2}$$

$$n + \frac{1}{4} \leq x \leq n + \frac{1}{2}, \quad n \in \mathbb{I}$$

5. **Ans. (C)**

Sol.



Number of solution of $f(-|x|) = \frac{1}{2}$ is 4

6. **Ans. (A)**

Sol. $|\cos x| \in [0, 1]$

$$|\sin x| \in [0, 1]$$

$$f(x)_{\min} = 4 \times 0 - 3 \times 1 = -3 \quad \text{at } x = \frac{\pi}{2}$$

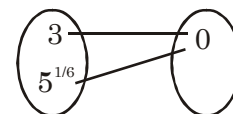
$$f(x)_{\max} = 4 \times 1 - 3 \times 0 = 4 \quad \text{at } x = 0$$

7. **Ans. (B,C)**

Sol. $f(3) = f(5^{1/6}) = 0$

Hence manyone

$$f(x) \neq -5^{1/2} \text{ Hence into}$$



8. **Ans. (A,C)**

Sol. $f(x) + f(-x) = \log(\sqrt{1+x^2} + x) + \log(\sqrt{1+x^2} - x)$

$$= \log(1 + x^2 - x^2) = 0, \text{ odd function}$$

$$(B) f(x) = x \left(\frac{e^x - 1}{e^x + 1} \right)$$

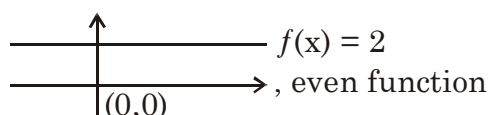
$$f(-x) = -x \left(\frac{e^{-x} - 1}{e^{-x} + 1} \right) = x \left(\frac{e^x - 1}{e^x + 1} \right) = f(x),$$

even function

$$(C) f(x) = (x^3 + x) |\sin x|$$

$$f(-x) = -(x^3 + x) |\sin x| = -f(x), \text{ odd function}$$

$$(D) f(x) = 2$$



9. Ans. (A,C)

Sol. $f(x) = x^2 - 4x + 5$

$$f'(x) = 2x - 4$$

$$(A) X = [2, \infty), Y = [f(2), \infty) \Rightarrow Y = [1, \infty)$$

$$X = (-\infty, 1), Y = (f(1), f(\infty)) \Rightarrow Y \in (2, \infty)$$

10. Ans. (A,D)

Sol. $[\sin 6x] : T_1 = \frac{\pi}{3}$

$$|\cos 12x| : T_2 = \frac{\pi}{12}$$

$$\text{LCM} \left\{ \frac{\pi}{3}, \frac{\pi}{12} \right\} = \frac{\pi}{3}$$

11. Ans. (A,B,C,D)

Sol. (A) $f(x) = |x|$, $g(x) = x$ NOT Identical

$$(B) f(x) = \sqrt{\frac{x-2}{x-3}}, g(x) = \frac{\sqrt{x-2}}{\sqrt{x-3}}$$

NOT Identical

$$x \in (-\infty, 2) \cup (3, \infty), x \in (3, \infty)$$

$$(C) f(x) = \tan x, g(x) = \frac{1}{\cot x}$$

NOT Identical

$$x \in \mathbb{R} - (2n+1)\frac{\pi}{2}, x \in \mathbb{R} - \left\{ n\pi, (2n+1)\frac{\pi}{2} \right\}$$

$$(D) f(x) = \begin{cases} 1 & x > 0 \\ -1 & x < 0 \end{cases}, g(x) = \text{sgn} x$$

NOT Identical

12. Ans. (B,C,D)

Sol. $\left| \frac{3x+4y-10}{5} \right| = \left| \frac{5x-12y-7}{13} \right|$

$$|39x + 52y - 130| = |25x - 60y - 35|$$

$$\text{+ve sign : } 14x + 112y - 95 = 0$$

$$\text{-ve sign : } 64x - 8y - 165 = 0$$

13. Ans. (A,C)

Sol. $(7x + y - 1) + \lambda(3x - 4y + 5) = 0$

$$(3\lambda + 7)x + (1 - 4\lambda)y + (5\lambda - 1) = 0$$

$$(A) \frac{3\lambda + 7}{4\lambda - 1} = -1 \Rightarrow \lambda = -\frac{6}{7}$$

$$x + y = \frac{37}{31}$$

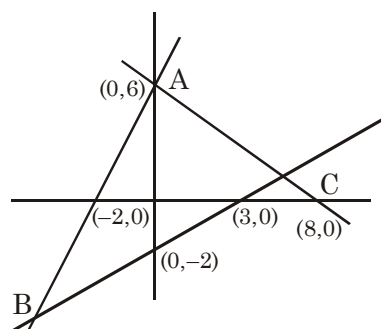
$$(B) \lambda = -\frac{7}{3}, y = \frac{38}{31}$$

$$(C) \xrightarrow{(1,1)} 3\lambda + 7 + 1 - 4\lambda + 5\lambda - 1 = 0$$

$$\Rightarrow \lambda = -\frac{7}{4}$$

14. Ans. (A,D)

Sol.



Paragraph for Question 15 to 16

15. Ans. (A)

Sol. $g(x) = \log_e(7x - x^2 - 10)$

$$7x - x^2 - 10 > 0 \Rightarrow x^2 - 7x + 10 < 0$$

$$x \in (2, 5)$$

16. Ans. (B)

Sol. $f(x) = -x^2 + 7x - 10$

$$f(x) \in \left(-\infty, \frac{9}{4}\right]$$

$$g(x) = \log_e f(x) \quad f(x) \in \left(-\infty, \frac{9}{4}\right]$$

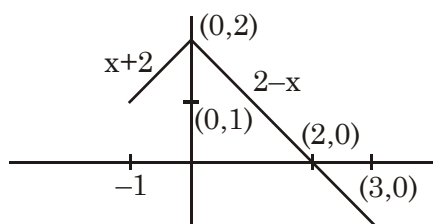
$$g(x) \in \left(-\infty, \log_e \frac{9}{4}\right]$$

Paragraph for Question 17 to 18

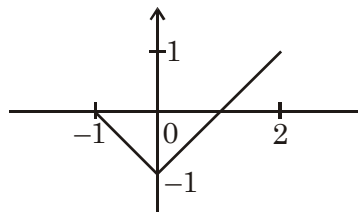
Sol. $f(x) = \begin{cases} 1-x & 0 \leq x \leq 2 \\ x-3 & 2 \leq x \leq 4 \end{cases}$

$$g(x) = \begin{cases} 2+x & -1 \leq x \leq 0 \\ 2-x & 0 \leq x \leq 3 \end{cases}$$

$$f(g(x)) = \begin{cases} 1-g(x) & 0 \leq g(x) \leq 2 \\ g(x)-3 & 2 \leq g(x) \leq 4 \end{cases}$$



$$y = f(g(x)) = \begin{cases} -(x+1) & -1 \leq x \leq 0 \\ (x-1) & 0 \leq x \leq 2 \end{cases}$$



$$y \in [-1, 1]$$

17. Ans. (C)

18. Ans. (D)